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(54) **SYSTEMS, APPARATUSES AND METHODS
FOR RADIO FREQUENCY-BASED
ATTACHMENT SENSING**

(71) Applicant: **Zoll Medical Israel Ltd.**, Kfar-Saba
(IL)

(72) Inventors: **Gil Arditi**, Binyamina (IL); **Roman
Vaistikh**, Ganei Tikva (IL); **Uriel
Weinstein**, Mazkeret Batya (IL)

(73) Assignee: **ZOLL Medical Israel Ltd.**, Kfar-Saba
(IL)

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A61B 5/00 (2006.01)

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(58) **Field of Classification Search**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,240,445 A 12/1980 Iskander et al.

4,344,440 A 8/1982 Aaby et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 101032400 A 9/2007

CN 101516437 A 8/2009

(Continued)

OTHER PUBLICATIONS

Alekseev, S. I., et al. "Human Skin permittivity determined by
millimeter wave reflection measurements", Bioelectromagnetics,
vol. 28, No. 5, Jul. 1, 2007, pp. 331-339.

(Continued)

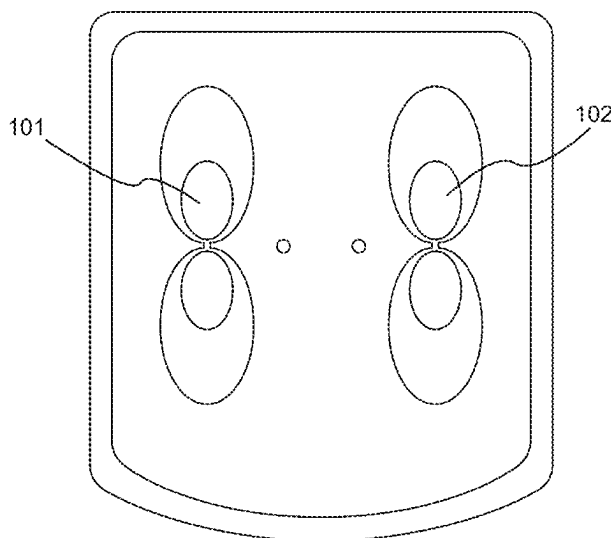
Primary Examiner — Nicole F Johnson

(74) *Attorney, Agent, or Firm* — The Webb Law Firm

(57) **ABSTRACT**

Some embodiments of the present disclosure discuss an
apparatus comprising a transceiver configured to generate
and/or receive radio frequency (RF) electromagnetic signals,
one or more antennae configured to radiate the generated RF
electromagnetic signals toward a surface and to output
signals corresponding to received reflections of the RF
electromagnetic signals, and a processing circuitry config-
ured to process the received reflections and/or the output
signals so as to determine change in position of the apparatus
with respect to the surface. The apparatus may be incorpo-
rated into a wearable garment and/or an adhesive patch, and
it may be attached to an outer surface of a human or an
animal body.

20 Claims, 4 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,557,272	A	12/1985	Carr	7,267,651	B2	9/2007	Nelson
4,632,128	A	12/1986	Paglione et al.	7,272,431	B2	9/2007	McGrath
4,640,280	A	2/1987	Sterzer	7,280,863	B2	10/2007	Shachar
4,641,659	A	2/1987	Sepponen	7,454,242	B2	11/2008	Fear et al.
4,774,961	A	10/1988	Carr	7,474,918	B2	1/2009	Frants et al.
4,825,880	A	5/1989	Stauffer et al.	7,479,790	B2	1/2009	Choi
4,926,868	A	5/1990	Larsen	7,493,154	B2	2/2009	Bonner et al.
4,945,914	A	8/1990	Allen	7,529,398	B2	5/2009	Zwirn et al.
4,958,638	A	9/1990	Sharpe	7,570,063	B2	8/2009	Van Veen et al.
4,986,870	A	1/1991	Frohlich	7,591,792	B2	9/2009	Bouton
5,003,622	A	3/1991	Ma et al.	7,697,972	B2	4/2010	Verard et al.
5,109,855	A	5/1992	Guner	7,719,280	B2	5/2010	Lagae et al.
5,394,882	A	3/1995	Mawhinney	7,747,302	B2	6/2010	Milledge et al.
5,404,877	A	4/1995	Nolan	7,868,627	B2	1/2011	Turkovskyi
5,474,574	A	12/1995	Payne et al.	8,032,211	B2	10/2011	Hashimshony et al.
5,540,727	A	7/1996	Tockman et al.	8,211,040	B2	7/2012	Kojima et al.
5,549,650	A	8/1996	Bornzin et al.	8,295,920	B2	10/2012	Bouton et al.
5,668,555	A	9/1997	Starr	8,352,015	B2	1/2013	Bernstein et al.
5,704,355	A	1/1998	Bridges	8,473,054	B2	6/2013	Pillai et al.
5,766,208	A	6/1998	McEwan	8,682,399	B2	3/2014	Rabu
5,807,257	A	9/1998	Bridges	8,882,759	B2	11/2014	Manley et al.
5,829,437	A	11/1998	Bridges	8,938,292	B2	1/2015	Hettrick et al.
5,841,288	A	11/1998	Meaney et al.	8,983,592	B2	3/2015	Belalcazar
5,865,177	A	2/1999	Segawa	8,989,837	B2	3/2015	Weinstein et al.
5,967,986	A	10/1999	Cimochowski et al.	9,220,420	B2	12/2015	Weinstein et al.
6,019,724	A	2/2000	Gronningsaeter et al.	9,265,438	B2	2/2016	Weinstein et al.
6,061,589	A	5/2000	Bridges et al.	9,572,512	B2	2/2017	Weinstein et al.
6,064,903	A	5/2000	Riechers et al.	9,629,561	B2	4/2017	Weinstein et al.
6,093,141	A	7/2000	Mosseri et al.	9,788,752	B2	10/2017	Weinstein et al.
6,144,344	A	11/2000	Kim	10,136,833	B2	11/2018	Weinstein et al.
6,161,036	A	12/2000	Matsumura et al.	10,548,485	B2	2/2020	Arditi et al.
6,193,669	B1	2/2001	Degany et al.	10,588,599	B2	3/2020	Weinstein et al.
6,208,286	B1	3/2001	Rostislavovich et al.	11,241,158	B2	2/2022	Arditi et al.
6,233,479	B1	5/2001	Haddad et al.	11,259,715	B2	3/2022	Weinstein et al.
6,267,723	B1	7/2001	Matsumura et al.	2002/0032386	A1	3/2002	Sackner et al.
6,330,479	B1	12/2001	Stauffer	2002/0045836	A1	4/2002	Alkawwas
6,409,662	B1	6/2002	Lloyd et al.	2002/0049394	A1	4/2002	Roy et al.
6,454,711	B1	9/2002	Haddad	2002/0050954	A1	5/2002	Jeong-Kun et al.
6,471,655	B1	10/2002	Baura	2002/0147405	A1	10/2002	Denker et al.
6,480,733	B1	11/2002	Turcott	2002/0151816	A1	10/2002	Rich et al.
6,526,318	B1	2/2003	Ansarinia	2003/0036674	A1	2/2003	Bouton
6,592,518	B2	7/2003	Denker et al.	2003/0036713	A1	2/2003	Bouton et al.
6,604,404	B2	8/2003	Paltiel et al.	2003/0088180	A1	5/2003	Van Veen et al.
6,729,336	B2	5/2004	Da Silva et al.	2003/0100815	A1	5/2003	Da Silva et al.
6,730,033	B2	5/2004	Yao et al.	2003/0199770	A1	10/2003	Chen et al.
6,755,856	B2	6/2004	Fierens et al.	2003/0219598	A1	11/2003	Sakurai
6,933,811	B2	8/2005	Enokihara et al.	2004/0015087	A1	1/2004	Boric-Lubecke et al.
6,940,457	B2	9/2005	Lee et al.	2004/0073081	A1	4/2004	Schramm
7,020,508	B2	3/2006	Stivoric et al.	2004/0077943	A1	4/2004	Meaney et al.
7,122,012	B2	10/2006	Bouton et al.	2004/0077952	A1	4/2004	Rafter et al.
7,130,681	B2	10/2006	Gebhardt et al.	2004/0249257	A1	12/2004	Tupin et al.
7,184,824	B2	2/2007	Hashimshony	2004/0254457	A1	12/2004	van der Weide
7,191,000	B2	3/2007	Zhu et al.	2004/0261721	A1	12/2004	Steger
7,197,356	B2	3/2007	Carr	2005/0038503	A1	2/2005	Greenhalgh et al.
7,266,407	B2	9/2007	Li et al.	2005/0107693	A1	5/2005	Fear et al.
				2005/0192488	A1	9/2005	Brynton
				2005/0245816	A1	11/2005	Candidus et al.
				2006/0004269	A9	1/2006	Caduff et al.
				2006/0009813	A1	1/2006	Taylor et al.
				2006/0025661	A1	2/2006	Sweeney et al.
				2006/0101917	A1	5/2006	Merkel
				2006/0237223	A1	10/2006	Chen et al.
				2006/0265034	A1	11/2006	Aknine et al.
				2007/0016032	A1	1/2007	Aknine
				2007/0016050	A1	1/2007	Moehring et al.
				2007/0055123	A1	3/2007	Takiguchi
				2007/0100385	A1	5/2007	Rawat
				2007/0123770	A1	5/2007	Bouton et al.
				2007/0123778	A1	5/2007	Kantorovich
				2007/0135721	A1	6/2007	Zdeblick
				2007/0152812	A1	7/2007	Wong et al.
				2007/0156057	A1	7/2007	Cho et al.
				2007/0162090	A1	7/2007	Penner
				2007/0191733	A1	8/2007	Gianchandani et al.
				2007/0263907	A1	11/2007	McMakin et al.
				2008/0027313	A1	1/2008	Shachar
				2008/0030284	A1	2/2008	Tanaka et al.
				2008/0036668	A1	2/2008	White et al.
				2008/0097199	A1	4/2008	Mullen

(56)	References Cited		2014/0081159 A1	3/2014	Tao et al.
	U.S. PATENT DOCUMENTS		2014/0128032 A1	5/2014	Muthukumar
			2014/0154991 A1*	6/2014	Brown G01F 1/66 455/42
2008/0129511 A1	6/2008	Yuen et al.	2014/0163425 A1	6/2014	Tran
2008/0139934 A1	6/2008	McMorrow et al.	2014/0288436 A1	9/2014	Venkatraman et al.
2008/0167566 A1	7/2008	Unver et al.	2015/0025333 A1	1/2015	Weinstein et al.
2008/0169961 A1	7/2008	Steinway et al.	2015/0150477 A1	6/2015	Weinstein et al.
2008/0183247 A1	7/2008	Harding	2015/0164349 A1	6/2015	Gopalakrishnan et al.
2008/0200802 A1	8/2008	Bahavaraju et al.	2015/0335310 A1	11/2015	Bernstein et al.
2008/0224688 A1	9/2008	Rubinsky et al.	2016/0073924 A1	3/2016	Weinstein et al.
2008/0269589 A1	10/2008	Thijs et al.	2016/0198957 A1	7/2016	Arditi et al.
2008/0283282 A1	11/2008	Kawasaki et al.	2016/0198976 A1	7/2016	Weinstein et al.
2008/0294036 A1	11/2008	Hoi et al.	2016/0213321 A1	7/2016	Weinstein et al.
2008/0316124 A1	12/2008	Hook	2016/0317054 A1	11/2016	Weinstein et al.
2008/0319301 A1	12/2008	Busse	2016/0345845 A1	12/2016	Ravid et al.
2009/0021720 A1	1/2009	Hecker	2017/0035327 A1	2/2017	Yuen et al.
2009/0048500 A1	2/2009	Corn	2017/0135598 A1	5/2017	Weinstein et al.
2009/0076350 A1	3/2009	Bly et al.	2017/0238966 A1	8/2017	Weinstein et al.
2009/0153412 A1	6/2009	Chiang et al.	2017/0296093 A1	10/2017	Weinstein et al.
2009/0153433 A1	6/2009	Nagai et al.	2019/0046038 A1	2/2019	Weinstein et al.
2009/0187109 A1	7/2009	Hashimshony	2019/0298208 A1	10/2019	Weinstein et al.
2009/0203972 A1	8/2009	Heneghan et al.	2020/0113447 A1	4/2020	Arditi et al.
2009/0227882 A1	9/2009	Foo	2022/0013899 A1	1/2022	Weinstein et al.
2009/0240132 A1	9/2009	Friedman			
2009/0240133 A1	9/2009	Friedman			
2009/0248450 A1	10/2009	Fernandez			
2009/0262028 A1	10/2009	Mumbru et al.			
2009/0281412 A1	11/2009	Boyden et al.			
2009/0299175 A1	12/2009	Bernstein et al.			
2009/0312615 A1	12/2009	Caduff et al.			
2009/0322636 A1	12/2009	Brigham et al.			
2010/0004517 A1	1/2010	Bryenton			
2010/0013318 A1	1/2010	Iguchi et al.			
2010/0052992 A1	3/2010	Okamura et al.			
2010/0056907 A1	3/2010	Rappaport et al.			
2010/0076315 A1	3/2010	Erkamp et al.			
2010/0081895 A1	4/2010	Zand			
2010/0106223 A1	4/2010	Grevious			
2010/0152600 A1	6/2010	Droitcour et al.			
2010/0256462 A1	10/2010	Rappaport et al.			
2010/0265159 A1	10/2010	Ando et al.			
2010/0305460 A1	12/2010	Pinter et al.			
2010/0312301 A1	12/2010	Stahmann			
2010/0321253 A1	12/2010	Ayala Vazquez et al.			
2010/0332173 A1	12/2010	Watson et al.			
2011/0004076 A1	1/2011	Janna et al.			
2011/0009754 A1	1/2011	Wenzel et al.			
2011/0022325 A1	1/2011	Craddock et al.			
2011/0040176 A1	2/2011	Razansky et al.			
2011/0060215 A1	3/2011	Tupin et al.			
2011/0068995 A1	3/2011	Baliarda et al.			
2011/0125207 A1	5/2011	Nabutovsky et al.			
2011/0130800 A1	6/2011	Weinstein et al.			
2011/0160802 A1*	6/2011	Rofougaran G16H 10/65 607/60			
2011/0257555 A1	10/2011	Banet et al.			
2012/0029323 A1	2/2012	Zhao			
2012/0065514 A1	3/2012	Naghavi et al.			
2012/0068906 A1	3/2012	Asher et al.			
2012/0098706 A1	4/2012	Lin et al.			
2012/0104103 A1	5/2012	Manzi			
2012/0242501 A1*	9/2012	Tran A61B 5/0024 340/870.02			
2012/0330151 A1	12/2012	Weinstein et al.			
2013/0041268 A1	2/2013	Rimoldi et al.			
2013/0053671 A1	2/2013	Farra			
2013/0069780 A1	3/2013	Tran et al.			
2013/0090566 A1	4/2013	Muhlsteff et al.			
2013/0123614 A1	5/2013	Bernstein et al.			
2013/0178958 A1*	7/2013	Kulach A63B 24/0021 700/91			
2013/0184573 A1	7/2013	Pahlevan et al.			
2013/0190646 A1	7/2013	Weinstein et al.			
2013/0225989 A1	8/2013	Saroka et al.			
2013/0231550 A1	9/2013	Weinstein et al.			
2013/0297344 A1	11/2013	Cosentino et al.			
2013/0310700 A1	11/2013	Wiard et al.			
2014/0046690 A1	2/2014	Gunderson et al.			
FOREIGN PATENT DOCUMENTS					
	DE	10008886		9/2001	
	EP	1834588 A1		9/2007	
	EP	2506917 A1		10/2012	
	EP	2 602 870 A1		6/2013	
	JP	05-038957		5/1993	
	JP	10-137193 A		5/1998	
	JP	2000-235006 A		8/2000	
	JP	2001-525925 A		12/2001	
	JP	2002-094321		3/2002	
	JP	2003-141466		5/2003	
	JP	2004-526488 A		9/2004	
	JP	2006-208070 A		8/2006	
	JP	2006-319767 A		11/2006	
	JP	2007-061359 A		3/2007	
	JP	2007-149959		6/2007	
	JP	2008-515548 A		5/2008	
	JP	2008-148141 A		6/2008	
	JP	2008-518706 A		6/2008	
	JP	2008-530546 A		7/2008	
	JP	2008-542759 A		11/2008	
	JP	2008-545471		12/2008	
	JP	2009-514619 A		4/2009	
	JP	2009-522034 A		6/2009	
	JP	2010-507929		3/2010	
	JP	2010-072957		4/2010	
	JP	2010-512190 A		4/2010	
	JP	2010-530769		9/2010	
	JP	2010-537766 A		12/2010	
	JP	2011-507583 A		3/2011	
	JP	2011-524213 A		9/2011	
	JP	2012-090257		5/2012	
	WO	WO 02/03499 A1		1/2002	
	WO	WO 2003/009752 A2		2/2003	
	WO	WO 2006/127719 A2		11/2006	
	WO	WO 2006/130798 A2		12/2006	
	WO	WO 2007/017861 A2		2/2007	
	WO	WO 2007/023426 A2		3/2007	
	WO	WO 2008/070856 A2		6/2008	
	WO	WO 2008/148040 A1		12/2008	
	WO	WO 2009/031149 A2		3/2009	
	WO	WO 2009/031150 A2		3/2009	
	WO	WO 2009/060182 A1		5/2009	
	WO	WO 2009/081331 A1		7/2009	
	WO	WO 2009/152625 A1		12/2009	
	WO	WO 2011/067623 A1		6/2011	
	WO	WO 2011/067685 A1		6/2011	
	WO	WO 2011/141915 A2		11/2011	
	WO	WO 2012/011065 A1		1/2012	
	WO	WO 2012/011066 A1		1/2012	
	WO	WO 2013/118121 A1		8/2013	

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO WO 2013/121290 A2 8/2013
 WO WO 2015/118544 A1 8/2015

OTHER PUBLICATIONS

Ascension Technology Corporation, "TrakStar Adds Versatility to Ascension's New Product Line: Desktop Model Joins driveBAY Tracker for Fast Guidance of Miniaturized Sensor", USA, Apr. 7, 2008.

Bell et al., "A Low-Profile Achimedean Spiral Antenna Using an EBG Ground Plane", IEEE Antennas and Wireless Propagation Letters 3, pp. 223-226 (2004).

Beyer-Enke et al., Intra-arterial Doppler flowmetry in the superficial femoral artery following angioplasty., 2000, European Radiology, vol. 10, No. 4, p. 642-649.

Claron Technology Inc., "Micron Tracker 3: A New Generation of Optical Trackers", Canada, 2009.

Czum et al., "The Vascular Diagnostic Laboratory", The Heart & Vascular Institute Newsletter, vol. 1, USA, Winter, 2001.

Extended Search Report for European Application No. 11809360.8, date of mailing, Mar. 11, 2014.

Ghosh, et al., Immediate Evaluation of Angioplasty and Stenting Results in Supra-Aortic Arteries by Use of a Doppler-Tipped Guidewire, Aug. 2004, American Journal of Neuroradiology, vol. 25, p. 1172-1176.

Gentili et al., "A Versatile Microwave Plethysmograph for the Monitoring of Physiological Parameters", IEEE Transactions on Biomedical Engineering, IEEE Service Center, Piscataway, NJ, US, vol. 49, No. 10, Oct. 1, 2002.

Haude et al., Intracoronary Doppler-and Quantitative Coronary Angiography-Derived Predictors of Major Adverse Cardiac Events After Stent Implantation, Mar. 6, 2001, Circulation, vol. 103(9), p. 1212-1217.

Immersion Corporation, "Immersion Introduces New 3D Digitizing Product-MicroScribe G2; Faster Data Transfer, USB Compatibility, New Industrial Design", Press Release, San Jose, USA, Jul. 1, 2002.

International Preliminary Report on Patentability, mailed Jan. 31, 2013, for International Application No. PCT/IB2011/053246, 22 pages.

International Preliminary Report on Patentability, dated Aug. 19, 2014 for International Application No. PCT/IB2013/000663 filed Feb. 15, 2013.

International Preliminary Report on Patentability, dated Jun. 5, 2012, for International Application No. PCT/IB2010/054861.

Kantarci et al., Follow-Up of Extracranial Vertebral Artery Stents with Doppler Sonography., Sep. 2006, American Journal of Roentgenology, vol. 187, p. 779-787.

Lal et al., "Duplex ultrasound velocity criteria for the stented carotid artery", Journal of Vascular Surgery, vol. 47, No. 1, pp. 63-73, Jan. 2008.

Larsson et al., "State Diagrams of the Heart a New Approach to Describing Cardiac Mechanics", Cardiovascular Ultrasound 7:22 (2009).

Liang, Jing et al., Microstrip Patch Antennas on Tunable Electromagnetic Band-Gap Substrates, IEEE Transactions on Antennas and Propagation, vol. 57, No. 6, Jun. 2009.

Lin, J.C. et al., "Microwave Imaging of Cerebral Edema", Proceedings of the IEEE, IEEE, NY, US, vol. 70, No. 5, May 1, 1982, pp. 523-524.

Lin et al., "Using dual-antenna nanosecond pulse near field sensing technology for non-contact and continuous blood pressure measurement", Engineering in Medicine and Biology Society (EMBC), 2013 35th Annual International Conference of the IEEE, IEEE, Aug. 28, 2012 (Aug. 28, 2012), pp. 219-222.

Miura et al., "Time Domain Reflectometry: Measurement of Free Water in Normal Lung and Pulmonary Edema," American Journal of Physiology—Lung Physiology 276:1 (1999), pp. L207-L212.

Paulson, Christine N., et al., "Ultra-wideband radar methods and techniques of medical sensing and imaging" Proceedings of SPIE, vol. 6007, Nov. 9, 2005, p. 60070L.

Pedersen, P.C., et al., "Microwave Reflection and Transmission Measurements for Pulmonary Diagnosis and Monitoring", IEEE Transactions on Biomedical Engineering, IEEE Service Center, Piscataway, NJ, US, vol. BME-19, No. 1, Jan. 1, 1978; pp. 40-48.

Polhemus, "Fastrak: The Fast and Easy Digital Tracker", USA, 2008.

Ringer et al., Follow-up of Stented Carotid Arteries by Doppler Ultrasound, Sep. 2002, Neurosurgery, vol. 51, No. 3, p. 639-643.

Solberg et al., "A feasibility study on aortic pressure estimation using UWB radar", Ultra-Wideband, 2009. ICUWB 2009. IEEE International Conference on, IEEE, Piscataway, NJ, USA, Sep. 9, 2009 (Sep. 9, 2009), pp. 464-468.

Yang, F. et al., "Enhancement of Printed Dipole Antennas Characteristics Using Semi-EBG Ground Plane", Journal of Electromagnetic Waves and Application, U.S., Taylor & Francis, Apr. 3, 2006, vol. 8, pp. 993-1006.

* cited by examiner

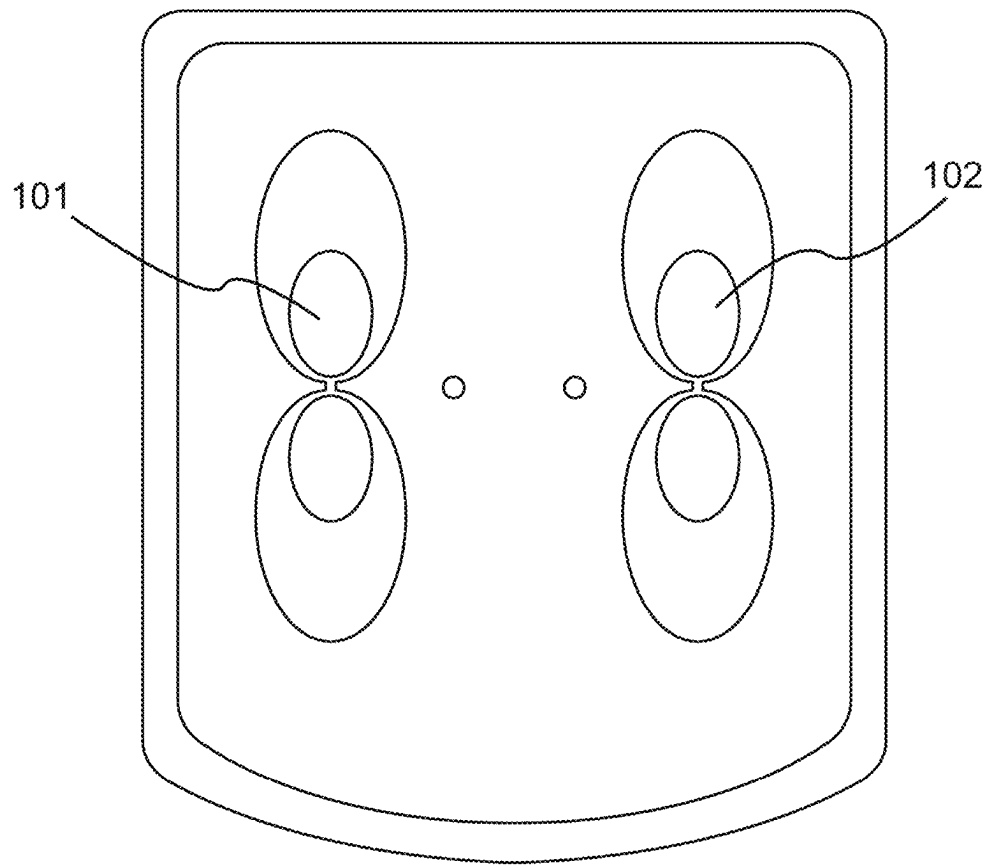


FIG. 1

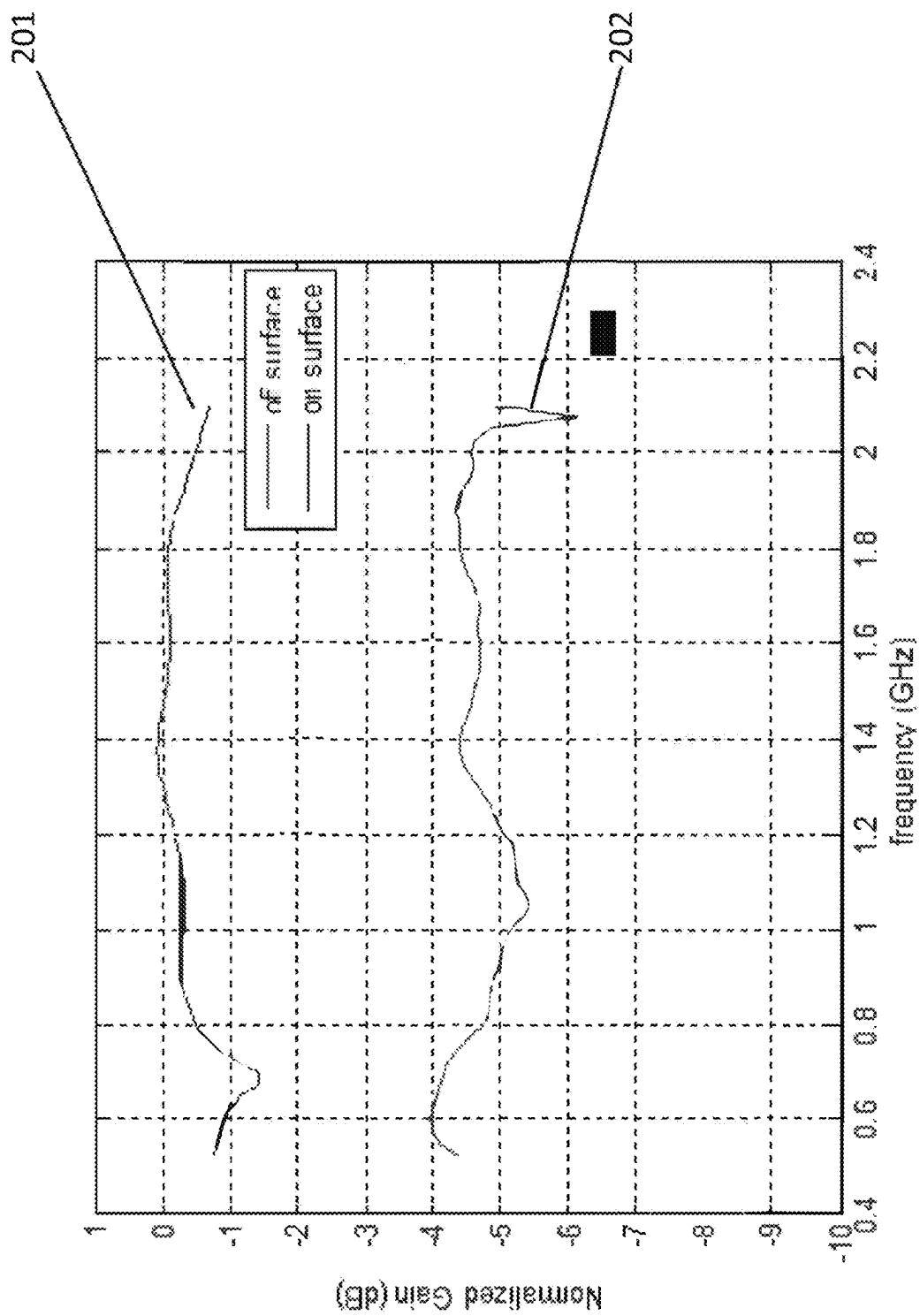


FIG. 2

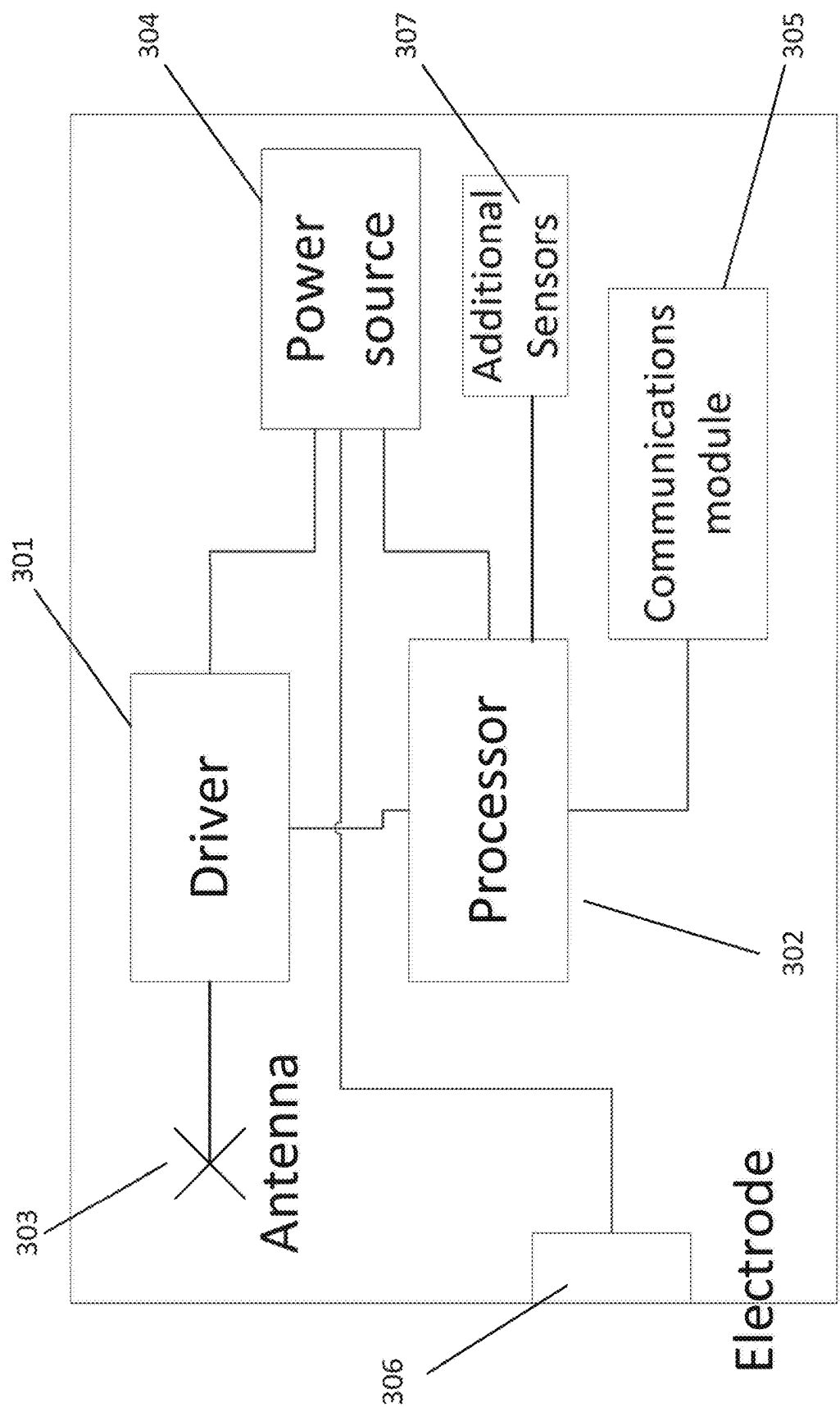


FIG. 3

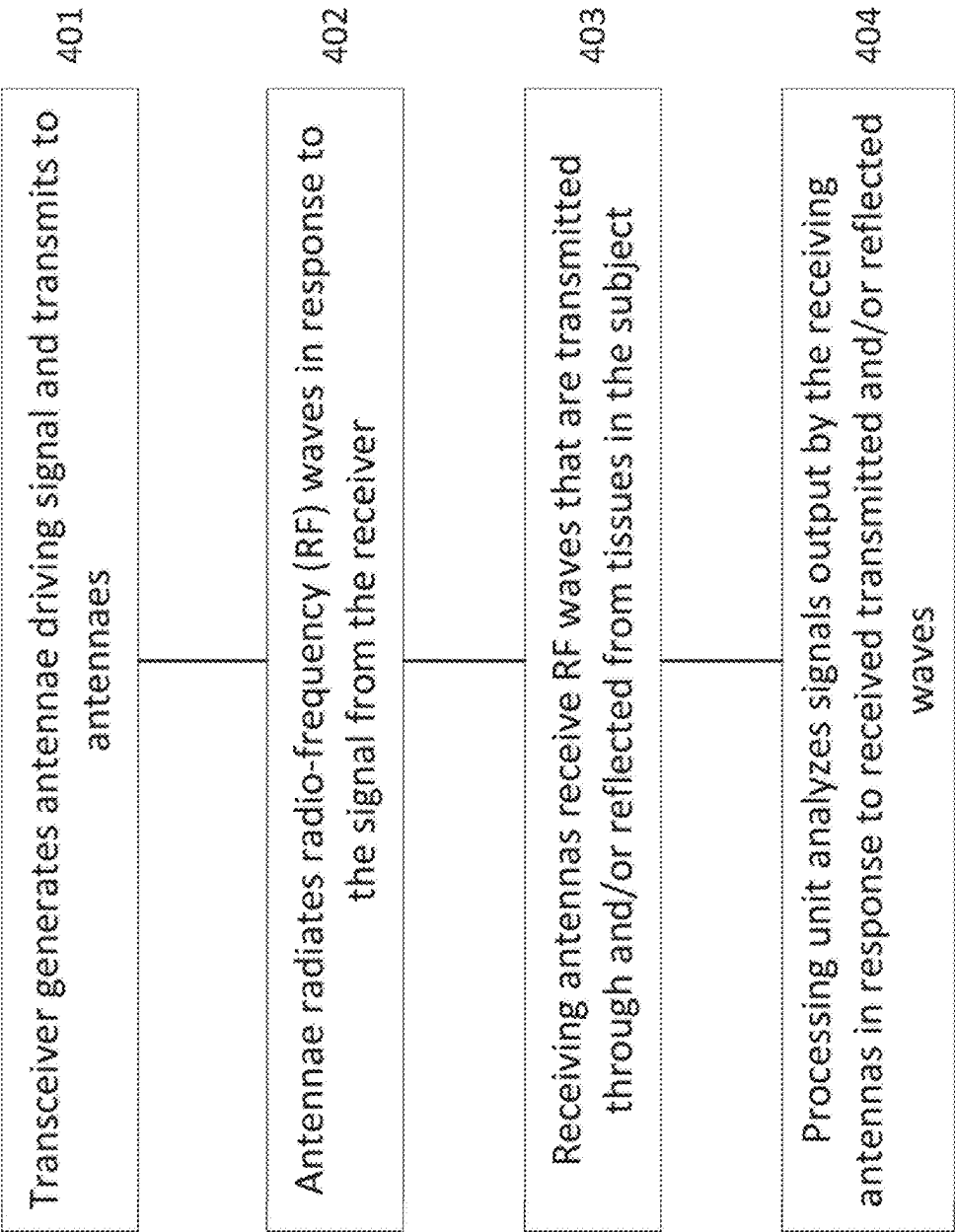


FIG. 4

SYSTEMS, APPARATUSES AND METHODS FOR RADIO FREQUENCY-BASED ATTACHMENT SENSING

CROSS REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/715,919, filed Dec. 16, 2019, entitled “Systems, Apparatuses and Methods for Radio Frequency-Based Attachment Sensing”, which is a continuation of U.S. patent application Ser. No. 14/994,052, filed Jan. 12, 2016, entitled “System, Apparatuses and Methods for Radio Frequency-Based Attachment Sensing”, and issued as U.S. Pat. No. 10,548,485, on Feb. 4, 2020, which claims benefit of and priority to U.S. Provisional Patent Application No. 62/102,551, entitled, “Systems, Apparatuses and Methods for Radio Frequency-based Attachment Sensing,” filed Jan. 12, 2015. Each of the foregoing disclosures is incorporated herein by reference in its entirety.

BACKGROUND

Radio-frequency (RF) electromagnetic radiation has been used for diagnosis and imaging of body tissues, examples of which may be found in PCT Patent Publication No. WO2011/067623, US Patent Publication Nos. 2009/0299175 and 2009/0240133, and U.S. Pat. Nos. 4,926,868, 5,766,208 and 6,061,589. Each of those disclosures is herein incorporated by reference in its entirety.

SUMMARY OF SOME OF THE EMBODIMENTS

Embodiments of antenna implementations discuss an apparatus comprising a transceiver configured to generate and/or receive radio frequency (RF) electromagnetic signals; one or more antennae configured to radiate the generated RF electromagnetic signals toward a surface and to output signals corresponding to received reflections of the RF electromagnetic signals; and a processing circuitry configured to process the received reflections and/or the output signals so as to determine change in position of the apparatus with respect to the surface. The apparatus may be incorporated into a wearable garment and/or an adhesive patch, and/or attached to an outer surface of a human body or an animal body. In some embodiments, the received RF electromagnetic signals may be reflected from the outer surface of a human or an animal body.

In some instances, the RF electromagnetic signals range in frequency from about 300 MHz to about 3 GHz. In some instances, the signals may range in frequency from about 300 MHz to about 300 GHz. In some instances, the received RF electromagnetic signals may be reflected from the surface.

In some embodiments, the one or more antennae may comprise a monostatic antenna element, a bistatic antenna element, and/or a polystatic antenna element. Further, the one or more antennae may comprise a flexible plane configured to conform to the surface, wherein the flexible plane comprises a printed circuit board.

In some embodiments, the apparatus further comprises an electrode configured to measure electrocardiographic signals of the human body or the animal body so as to determine attachment level of the apparatus to the human body or the animal body. It may also comprise a wireless communication unit or module configured to wirelessly communicate with an external module. In some instances, it

may further contain an alarm unit configured to provide a warning when the determination of the change in position of the apparatus with respect to the surface indicates attachment of the apparatus to the surface is below a predetermined attachment level threshold.

In some embodiments, the apparatus is configured to detect surface movements that are indicative of physiological effects, examples of which include heartbeat, muscle movement, and respiration. In some instances, the apparatus may be connected to a device for measuring physiological parameters, and may further comprise a switch unit configured to activate or deactivate the device based on the determination of the change in position of the apparatus with respect to the surface. In such implementations, the determination of the change in position of the apparatus with respect to the surface may indicate attachment of the apparatus to the surface is below a predetermined attachment level threshold. In some embodiments, the processing of the generated and/or the received signals comprises analyzing amplitudes and/or phases of the generated and/or the received signals. For example, the analysis of the amplitudes and/or the phases of the generated and/or the received signals comprises comparing changes in amplitudes and/or phases between the generated and the received signals to respective changes in amplitudes and/or phases recorded when the apparatus is at rest with respect to the surface. In some embodiments, such analysis may include determining signal delays of the generated and/or the received signals.

In some embodiments, a method for sensing change in position of an apparatus with respect to a surface to which the apparatus is secured to is disclosed. Such a method may comprise the step of generating and receiving, by a transceiver, radio frequency (RF) electromagnetic signals. It may also include radiating, by one or more antennae, the generated RF electromagnetic signals into the surface the apparatus is secured to; and outputting signals by the one or more antennae and in response to received reflected RF electromagnetic signals. In addition, it may comprise processing the received and/or the output signals so as to determine change in position of the apparatus with respect to the surface.

In some embodiments, a method for sensing attachment level of an apparatus with respect to a body surface to which the apparatus is attached to is disclosed. The method comprises generating and receiving, by a transceiver, radio frequency (RF) electromagnetic signals; and radiating, by one or more antennae, RF electromagnetic signals generated by a transceiver onto the body surface the apparatus is attached to. In some instances, the method includes outputting signals by the one or more antennae in response to RF electromagnetic signals reflected by the body surface; and analyzing the output signals with respect to a nominal signal to determine a shift in the attachment of the apparatus to the body surface, wherein the nominal signal is detected when the apparatus is securely attached to the body surface. For example, analyzing the output signals with respect to the nominal signal comprises calculating sum of absolute values of differences of amplitudes of the reflected signal from amplitudes of the nominal signal per frequency.

It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail below (provided such concepts are not mutually inconsistent) are contemplated as being part of the inventive subject matter disclosed herein. In particular, all combinations of claimed subject matter appearing at the end of this disclosure are contemplated as being part of the inventive subject matter disclosed herein. It should also be

appreciated that terminology explicitly employed herein that also may appear in any disclosure incorporated by reference should be accorded a meaning most consistent with the particular concepts disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

The skilled artisan will understand that the drawings primarily are for illustrative purposes and are not intended to limit the scope of the inventive subject matter described herein. The drawings are not necessarily to scale; in some instances, various aspects of the inventive subject matter disclosed herein may be shown exaggerated or enlarged in the drawings to facilitate an understanding of different features. In the drawings, like reference characters generally refer to like features (e.g., functionally similar and/or structurally similar elements).

FIG. 1 shows a picture of an example bistatic antenna for sensing attachment to a surface.

FIG. 2 illustrates an example normalized transmission loss (S21 parameter) comparison between embodiments when the antenna is attached to a surface, and when it is not.

FIG. 3 shows a schematic block diagram comprising functional components of an apparatus for processing signals to determine attachment level of the apparatus to a surface.

FIG. 4 is an example flow diagram illustrating the use of the disclosed apparatus for sensing changes in attachment of the apparatus to a surface of a human or animal body, according to an embodiment.

DETAILED DESCRIPTION OF SOME OF THE EMBODIMENTS

In some embodiments of the present disclosure, systems and apparatuses for radio frequency (RF)-based surface attachment sensing are presented. For example, in some embodiments, a sensing apparatus for sensing change in position of the apparatus with respect to a surface comprises one or more antennae, a transceiver, and a processing circuitry. In some embodiments, the sensing apparatus may comprise one or more antennae and a transceiver, and the apparatus may be operationally coupled to the processing circuitry. In some instances, the apparatus may be in direct contact with the surface (with or without adhesives, for example), and sensing and/or measuring the shift of the apparatus with respect to the surface may indicate the attachment level of the apparatus to the surface. In some instances, the apparatus may not be in direct contact to the surface but secured or stably connected to it. For example, the apparatus, while not directly contacting the surface, may be incorporated into a garment or patch that is secured to the surface. In some embodiments, the surface may be an outer surface of a human or an animal body, such as the skin.

In some embodiments, the transceiver can generate driving signals for transmission by the one or more antenna, and may further filter and interpret reflected and propagated signals that are received by the antennae. The transceiver may generate the signals at multiple different frequencies, and when signals are received by the antennae, the received signals may be digitized by the transceiver. In some instances, some or all of the signals may be reflected by the surface and/or body tissues along the path of the signals. For example, RF signals generated by a transceiver and radiated by an antenna of an apparatus attached to the surface of a human or animal body may be reflected by the skin, scattered by body tissues along the propagation path, and/or

emerge on substantially opposite side of the body as propagated waves. In some instances, the apparatus may comprise additional antennae on the opposite side of the body to receive the propagated waves.

In some embodiments, the antenna comprises a single antenna element (monostatic antenna), a bistatic antennae (two antenna elements), and/or polystatic or multistatic antennae (more than two antennae elements). In a monostatic mode, the antenna may switch between transmitting and receiving modes, and radiate RF signals and receive reflected waves back from the surface the apparatus is attached to, and/or other tissues on the path of the radiated wave. In bistatic and polystatic modes, antenna elements may comprise a plurality of antenna elements, each tasked with transmitting or receiving RF signals. For example, a bistatic antenna may have one antenna element radiating RF signals and another element receiving waves that are reflected as well as waves radiated by other antenna elements. In some embodiments, the RF signals waveform can be continuous or based on step frequency, and may be radiated in the frequency range of from about 200 MHz to about 300 GHz. The RF signals may also be radiated in the frequency range from about 250 MHz to about 100 GHz, from about 300 MHz to about 30 GHz, from about 300 MHz to about 3 GHz, from about 500 MHz to about 1 GHz, about 300 MHz, about 3 GHz, and/or the like. In yet some instances, higher and lower frequencies outside these ranges may be used. In some of these embodiments, the various frequency elements may be used to improve the performance of the apparatus. For example, multiple frequencies (e.g., from any of the above-noted frequency ranges) may be used to improve detection. Modulation of the frequencies may allow reduction or elimination of interferences. The RF signals may also be configured so as to achieve a range (e.g., depth) resolution that allows effective filtering of reflections from different depths of the surface of the body.

Various types of antennas may be used in implementing embodiments of the present disclosure, including wire antennas (e.g., monopole, dipole, loop antennas, etc.), microstrip antennas (e.g., patch, planar), aperture antennas (e.g., slot, cavity-backed, planar), traveling wave antennas (e.g., helical), printed antennas and/or the like. For example, planar antenna backed by an in-phase reflective structure based on an electromagnetic band gap (EBG) structure between an antenna ground plane and the front surface may be used. Such structures are especially advantageous for providing flat, and possibly flexible antenna, that may conform to a body surface. In some instances, the antennae may comprise a flexible plane attached to said surface. For example, the antenna may be printed on a flexible printed circuit board (PCB). In some instances, planar antenna comprising a conductive element (e.g., spiral) on the front surface of the antenna may be used. Conductive elements in such antennae may be able to receive electrocardiogram (ECG) signals from the body surface, obviating the need for separate ECG electrodes. In some instances, the conductive elements may be included in printed circuit boards that the antennae comprise on the front surface of the antennae.

In some embodiments, the apparatus may comprise a processing circuitry that includes a processor capable of processing signals received by the one or more antennae and/or the transceiver to determine attachment level of the apparatus to the surface the signals are reflected from or propagated through. In some embodiments, the apparatus may not comprise the processing circuitry, but instead may be operationally coupled to the circuitry. In some instances, the processor accomplishes this task by analyzing the ampli-

tudes and/or phases of these signals. For example, the sensing apparatus may process the reflected signals to determine the change in the amplitude and/or phase of the reflected/propagated signals with respect to the signals that were radiated into the surface by the antennae. The processor may then compare these changes to benchmark changes to determine the attachment level of the apparatus to the surface. Examples of benchmark changes are changes in amplitudes and/or phases that have been previously recorded under known conditions. For example, a measurement for amplitude and/or phase changes may be made when the attachment level of the apparatus to the surface is known (e.g., the apparatus is not moving with respect to the surface, the apparatus is directly attached to the surface, the apparatus is in proximity to the surface within a known separation distance, the apparatus is incorporated into a wearable garment or adhesive patch, etc.). Such measurements may then be used as benchmarks for comparison with amplitude/phase changes that the processor determines by analyzing the reflected and/or propagated signals. In some embodiments, the sensing apparatus may process the reflected signals to determine signal delays of the generated and/or the reflected signals. For example, the signal delays may be obtained from an analysis of the phases of the signals.

In some embodiments, the apparatus may process the reflected signals to determine the change in the amplitude and/or phase of the reflected/propagated signals with respect to a nominal signal when the attachment level of the sensing apparatus to the surface is known (e.g., the nominal signal corresponds to when the apparatus is not moving with respect to the surface, the apparatus is directly attached to the surface, the apparatus is in proximity to the surface within a known separation distance, the apparatus is incorporated into a wearable garment or adhesive patch, etc.). The changes may then be used as measures of attachment levels. For example, a measure of attachment can be determined, in some embodiments, according to the following procedure. First, a nominal signal (including the amplitude thereof) is detected upon the apparatus being attached to a surface of a body (e.g., skin). RF signals are then transmitted from antenna and, reflections therefrom are received by the antenna (either the same antenna or another antenna). The reflected signals are then analyzed to determine the amplitudes of reflected signals with respect to the amplitudes of the nominal signal, for example, by calculating the sum of the absolute value of the difference of the reflected signal amplitude per frequency from a nominal amplitude set. The difference may represent the state of attachment.

In some embodiments, one may wish to check that the sensing apparatus has not been removed from the intended subject (usually a living being such as a human or animal), either intentionally or accidentally, so as to protect the integrity of the measurement results. For example, the apparatus may be attached to a non-living stationary surface and indicate undisturbed attachment, leading to misleading interpretations. In such situations, one may utilize an accelerometer to better determine if the subject the apparatus is intended to be attached to is in fact a living being or not, or at least capable of movement. For example, an accelerometer can be used to better distinguish between attachment to a stationary surface and attachment to living being or tissue (that is, a body capable of movement). In some embodiments, this can be performed by calculating the amount of accelerometer signal energy in low frequencies compared to the energy of the entire spectrum.

In some embodiments, the processing circuitry comprises a combination of dedicated hardware circuits and program-

mable components. Programmable processors, such as a general-purpose microprocessor, a digital signal processor, etc., which are programmed in software may be used to carry out the tasks described herein. In some implementations, the processing circuitry may comprise dedicated or programmable digital logic units such as an application-specific integrated circuit (ASIC) or a field-programmable gate array (FPGA). In some instances, the tasks may be carried out by a single processor, or by a combination of one or more processors.

In some embodiments, the processing unit may communicate with an external server or device via a communication unit or module that may or may not be a component of the sensing apparatus. For example, any of the measurements obtained and/or processed by the processing unit may be transmitted to an external device by the communication unit. In addition, the communication unit may be configured to receive any data, instructions, etc., from an external device.

The apparatus as described herein may be utilized for a variety of purposes, an example of which is to determine if devices used for measuring physiological parameters on a human or animal skin have maintained their contact with the skin. For example, devices or sensors such as an ECG sensor, photoplethysmogram (PPG) sensor, etc., may be securely attached to the sensing apparatus described herein, and the apparatus/sensor may be used to check on the attachment level of the apparatus itself (and hence the ECG, PPG devices) to the skin as the devices are taking measurements. In some embodiments, the apparatus may comprise an electrode or a conductive element that is configured to receive ECG, PPG, etc., signals from the surface the apparatus is attached to. In such embodiments, the apparatus itself may also serve as a physiological sensor (e.g., ECG and/or PPG devices). In such embodiments, the apparatus may use the measurements of a body as proxy measures of attachment levels of the apparatus to the surface of the body. For example, the processor of the processing unit contained by or operationally coupled to the apparatus may process the measured signals to compare those to benchmark or expected activities of the body. Deviations from the benchmark or expected activities may indicate that the attachment level of the apparatus is different from the attachment level when the benchmark or expected activities were observed. For example, a decreased intensity in the measured ECG or PPG activities may be interpreted as: the apparatus not being securely attached to the surface, the apparatus not being directly attached to the surface, the apparatus moving with respect to the surface, etc. In some embodiments, the apparatus may detect surface movements that are indicative of physiological effects, and these measurements may be used for, for example, diagnostic values. For example, the apparatus may detect minute movements of the surface the apparatus or the antennae are attached to (e.g., blood vessel, muscle, skin movements such as contractions and relaxations), and from the measurements, one may make diagnostic determinations about heartbeat, muscle movement, respiration, etc. Examples of muscle movements include movements of muscles of mastication, movements of muscles of swallowing, and/or the like.

In some embodiments, the devices may not necessarily be configured to receive signals from the body or tissue. For example, the devices may comprise electrodes that provide energy or current to the body such as, but not limited to, defibrillators that provide electrical current into the body. The electrodes may be configured to be attached to the outer surface of a human or an animal body, and the electrode may provide electrical current to the body.

In using the apparatus to measure attachment levels or even ECG/PPG activities, in some embodiments, it may be of interest to determine that the body of the surface to which the apparatus is attached to is a living body/tissue. For example, the attachment level measurements from the apparatus may indicate that the apparatus is firmly attached to the surface. However, the surface may not in fact be the surface of the body for which the attachment measurements are desired. For example, the surface may be some still surface that in fact may not be, for example, the skin of the body whose physiological parameters are being measured the apparatus or a sensor attached thereto (e.g., the sensor falls off a skin and attaches to a still surface, and upon measurement, this may lead to the inaccurate conclusion that the sensor is attached well to the skin). In some embodiments, the apparatus may comprise an accelerometer that detects movements of the apparatus, and by extension, devices and bodies securely attached to the apparatus. In some embodiments, the apparatus may comprise, or be incorporated into, other sensors/detectors such as but not limited to lie detectors, photoplethysmogram (PPG) sensors, ECG sensors, etc. In addition, the apparatus may be utilized to verify the efficacy of other surface attachment systems. For example, the apparatus can be incorporated into the surface attachment system of an insulin pump attachment so as to monitor and verify the effectiveness of the attachment system to the body to which the insulin pump is attached. Similarly, the apparatus can be incorporated into other attachment systems used to attach the aforementioned sensors (PPG sensors, ECG sensors, accelerometers, defibrillators, etc.) to a body. Further, the disclosed apparatus may be used with seatbelts where the apparatus may be used to verify and/or monitor whether seatbelts are properly in contact with users of the seatbelts. In some instances, this may provide additional information on the state of the body the apparatus is attached to (e.g., if the accelerometer indicates no movement on the part of the body for an extended period of time, then this may be an indication that the attachment may not have been to a living body or tissue).

In some embodiments, the apparatus disclosed herein may also be used to determine the locations of subcutaneous devices, examples of which include implants, subcutaneous devices, guiding sensors (e.g., guidance sensors used for subcutaneous needles, etc.). In some embodiments, the disclosed RF-based surface attachment sensing apparatus may direct RF signals towards locations of subcutaneous devices in a body, and based on an analysis of the reflection signals or patterns may determine the locations and/or change in locations of the devices. For example, changes in the amplitude phases, and/or signal delays of the reflected signals may be compared to benchmark values for indications on the locations and/or shifts in locations of the subcutaneous devices. In some embodiments, the verification or monitoring of the subcutaneous devices may be performed in real time, providing valuable feedback to a user of the disclosed apparatus (e.g., surgeon using a subcutaneous needle may use the disclosed apparatus to track the progress of the needle inside a patient's body).

In some embodiments, the apparatus or a sensor associated (e.g., securely connected) with the apparatus may be performing a physiological measurement of a body, and the apparatus may indicate that the attachment level of the apparatus/sensor is different from the appropriate level of attachment for performing the measurement. For example, the apparatus may comprise an alarm component that provides a warning when the attachment is below a predetermined attachment level threshold. The warning may be

visual (e.g., lights, textual displays, etc.), auditory (e.g., a warning via a speaker contained in the apparatus), tactile, and/or the like. In some instances, the apparatus may provide a value indicating the attachment level. For example, from analyzing the reflected and/or the propagated signals and/or measurements of ECG activities, the sensor may determine and assign a value to the attachment level (e.g., a 10% reduction in the intensity of ECG activities translates to an 85% attachment level based on a pre-determined calibration scheme).

Upon detecting that the attachment level is different from the level determined appropriate for the physiological measurement, in some embodiments, the apparatus may initiate a scheduling mechanism for the measurement. In some embodiments, the apparatus may comprise a switch that controls modes of operation of the sensor and/or devices connected to the apparatus. Based on the determination of attachment levels, the apparatus may activate or deactivate measurements being carried out (by the apparatus, additional sensors, etc.). In some embodiments, in addition to or instead of the ability to activate or deactivate measurements, the apparatus may have the capacity to alter the performance of the measurements according to the measured levels of attachment. For example, upon determining the attachment levels, the transceiver and/or the antennae may receive feedback to alter (e.g., increase or decrease) the radiation level if the apparatus determines the attachment to a surface has changed. In some instances, the apparatus may be monitoring the attachment level continuously or intermittently.

In some embodiments, attachment levels to a surface being different than their appropriate levels may mean excessive and/or long-lasting pressure is being applied to the surface of the body. For example, pressure sores or ulcers may develop if a body surface is exposed to a prolonged and/or excessive pressure, and the apparatus of the present disclosure may be used to monitor the pressure location and if needed provide alert as discussed above. In some embodiments, a compressible substance may be placed between the body surface and the apparatus, and RF signal reflections from the surface may be used in determining the level of the pressure (based on the attachment level, for example). In some embodiments, there may not be a compressible substance in between the surface and the apparatus, and reflections from layers below the surface may be used in determining attachment level and hence the pressure.

With reference to FIG. 1, in some embodiments, a picture of an example RF bistatic system (two antennas) for sensing attachment to a surface is shown. In some instances, one antenna, e.g., **101**, operates as transmission antenna and the second one, e.g., **102**, as receiving antenna. The antennae shown can be printed on printed circuit boards (PCBs), using a minimal form factor of bow-tie design, allowing wideband transmission and allowing for a wide range of dielectric constant material in the near field.

With reference to FIG. 2, in some embodiments, an example transmission loss (S21 parameter) comparison between embodiments when the antenna is attached to a surface **201**, and when it is not attached **202**, is shown. It can be seen that S21 parameters are significantly different when comparing the two states (attached to the surface **201**, and not attached **202**). In the example shown, the antenna elements were first placed against human skin and then slightly shifted away from it. Consequently, overall transmission gain has dropped by 3-5 dB over the entire tested spectrum. The opposite result can also be achieved if, for example, an antenna designed to be matched to air is placed

against a material different from air. In that case once the antenna is distanced from the material, transmission gain can increase significantly.

With reference to FIG. 3, in some embodiments, a schematic block diagram comprising functional components of an apparatus for processing signals to determine attachment level of the apparatus to a surface is shown. The apparatus comprises a processing circuitry including an antenna driver 301 and a processor 302. The antenna driver 301 drives one or more antennae 303 to emit RF waves onto the surface the apparatus is attached to. In some instances, the processor 302 processes signals that are output by the antennae 303 in response to waves reflected from the surface. The power source 304 supplies operating power to all the components of the apparatus, including to a communications module 305 used to relay data between the apparatus and an external device (e.g., computer). In some embodiments, electrodes 306 may be used to receive ECG signals from the body surface the apparatus is attached thereto. Further, additional sensors 307 such as accelerometers may be incorporated into the disclosed apparatus.

While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be an example and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure. Embodiments disclosed herein may also be combined with one or more features, as well as complete systems, devices and/or methods, to yield yet other embodiments and inventions. Moreover, some embodiments, may be distinguishable from the prior art by specifically lacking one and/or another feature disclosed in the particular prior art reference(s); i.e., claims to some embodiments may be distinguishable from the prior art by including one or more negative limitations.

Also, various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

Any and all references to publications or other documents, including but not limited to, patents, patent applications, articles, webpages, books, etc., presented anywhere in the present application, are herein incorporated by reference in their entirety. Moreover, all definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

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In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

What is claimed is:

1. An apparatus comprising:
 - a transceiver configured to generate and/or receive radio frequency (RF) electromagnetic signals;
 - one or more antennae configured for removable placement at a location on or immediately adjacent skin of a human or animal body and configured to:
 - radiate the generated RF electromagnetic signals into and through the skin, and
 - output signals corresponding to received reflections of the RF electromagnetic signals;
 - an accelerometer configured to detect attachment to a living being or tissue and output an accelerometer signal corresponding thereto;
 - an electrode configured to be attached to or contact the skin of the human body or the animal body; and
 - a processing circuitry configured to process the received reflections and/or the output signals, as well as to process the accelerometer signal so as to:
 - determine values corresponding to one or more parameters of the received reflections and/or the output signals,
 - determine diagnostic values indicative of at least one physiological effect based on the values corresponding to one or more parameters of the received reflections and/or the output signals, and
 - determine attachment of the apparatus to a living being or tissue based on the accelerometer signal and values corresponding to the one or more parameters of the received reflections and/or the output signals,
- wherein the parameters comprise amplitude and/or phase of the reflections and/or output signals, and
- wherein the electrode is configured to provide electrical current to the human body or the animal body.
2. The apparatus of claim 1, wherein determining attachment of the apparatus to a living being or tissue comprises calculating an amount of accelerometer signal energy in low frequencies compared to the energy of an entire spectrum.
3. The apparatus of claim 1, wherein determining attachment of the apparatus to a living being or tissue comprises determining whether the apparatus is attached to a stationary surface.
4. The apparatus of claim 1, wherein processing circuitry is configured to process the reflections and/or the output signals to determine amplitudes of reflected signals with respect to an amplitude of a nominal signal.
5. The apparatus of claim 4, wherein processing circuitry is configured to process the reflections and/or the output signals to determine the amplitudes of reflected signals with respect to the amplitude of the nominal signal by calculating a sum of an absolute value of a difference of the reflected signal amplitude per frequency from a nominal amplitude set.

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6. The apparatus of claim 1, wherein a waveform of the RF electromagnetic signals is continuous and/or based on step-frequency.

7. The apparatus of claim 1, wherein the RF electromagnetic signals range in frequency from about 300 MHz to about 300 GHz.

8. The apparatus of claim 1, wherein the one or more antennae comprise one or more of a monostatic antenna element, a bistatic antenna element, and a polystatic antenna element.

9. The apparatus of claim 1, wherein the one or more antennae comprise a flexible plane configured to conform to the skin of the human or animal body, wherein the flexible plane includes a printed circuit board.

10. The apparatus of claim 1, wherein the apparatus is incorporated into a wearable garment.

11. The apparatus of claim 1, wherein the electrode is configured to measure electrocardiogram and/or photoplethysmogram signals of the human body or the animal body, wherein the measurements are used to determine attachment level of the apparatus to the human body or the animal body.

12. The apparatus of claim 1, further comprising a wireless communication unit configured to wirelessly communicate with an external module.

13. The apparatus of claim 1, further comprising an alarm unit configured to provide a warning when a determination of a change in position of the apparatus with respect to the skin of the human or animal body indicates attachment of the apparatus to the skin of a human or animal body is below a predetermined attachment level threshold.

14. The apparatus of claim 1, wherein the apparatus is configured to detect surface movements that are indicative of physiological effects, the physiological effects including heartbeat, muscle movement, and respiration.

15. The apparatus of claim 14, wherein the muscle movement includes movement of muscles of mastication and/or movements of muscles of swallowing.

16. The apparatus of claim 1, wherein the apparatus is connected to a device for measuring physiological parameters, and further comprising a switch unit configured to activate or deactivate the device based on a determination of a change in position of the apparatus with respect to the skin of a human or animal body.

17. The apparatus of claim 1, wherein processing the generated and/or the received signals comprises determining signal delays of the generated and/or the received signals.

18. The apparatus of claim 1, wherein the at least one physiological effect relates to at least one of respiration, heartbeat, or muscle movement.

19. The apparatus of claim 1, wherein determining attachment of the apparatus to a living being or tissue comprises comparing the values corresponding the one or more parameters of the received reflections and/or the output signals with a range of known values.

20. The apparatus of claim 1, wherein determining attachment of the apparatus to a living being or tissue comprises comparing the values corresponding to the one or more parameters of the received reflections and/or the output signals with corresponding initial benchmark or nominal values, and determining attachment of the apparatus to a living being or tissue based on the comparison.

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